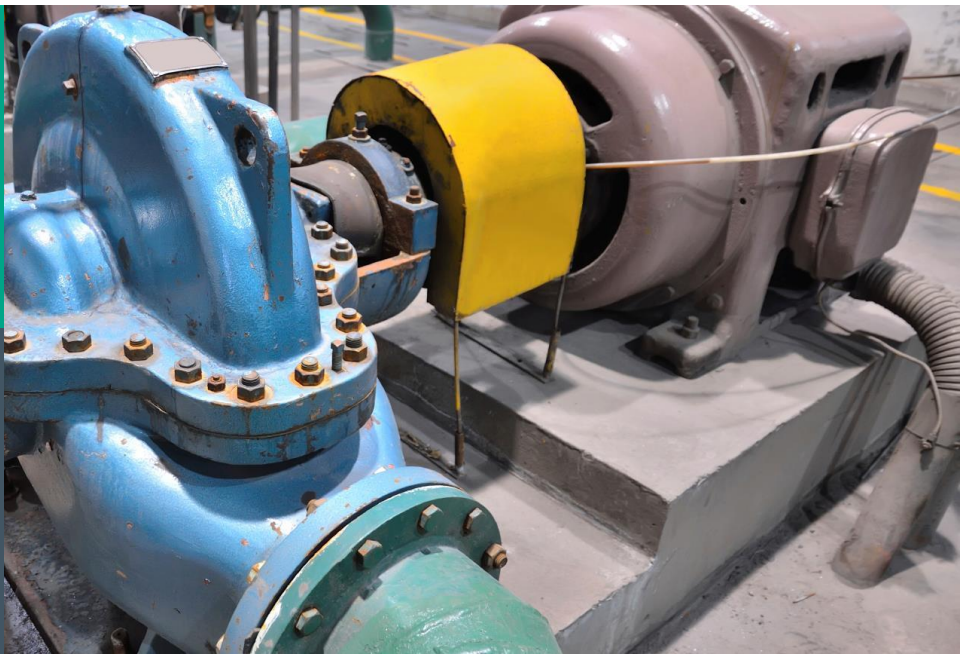
Explosion Protection

Non-Electrical Equipment for Hazardous Areas



Your automation, our passion.

PEPPERL+FUCHS

Overview

1. **Standards**

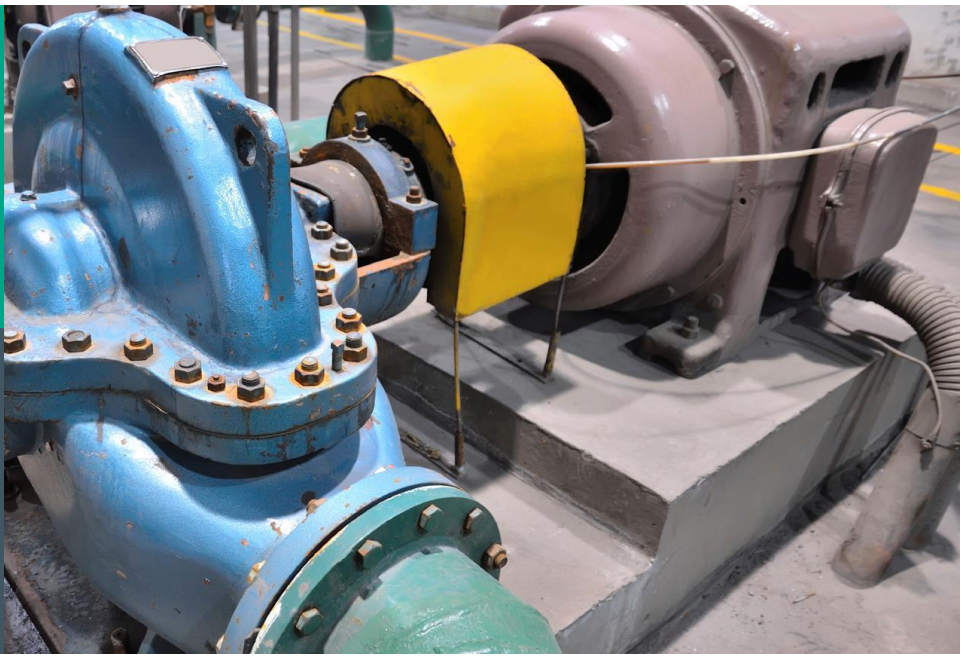
   EN ISO 80079-36 and -37, ignition hazard assessment

2. **Types of protection**

   c, b and k, functional principle, marking

1. Standards

EN ISO 80079-36 and -37, ignition hazard assessment



Standards: ISO EN 80079-36 and -37



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The series of standards ISO EN 80079-36 and -37 is applicable for:

- Design, construction, testing and marking of non-electrical equipment
- Hazardous areas: gas, steam, mist, dust
- Equipment groups I, II and III of all categories
  - → *Equipment standard* *(previously: EN 13463-...)*

Overview: Series of Standards ISO EN 80079

Series of standards "Non-electrical equipment for use in potentially explosive atmospheres" consists of:

- ...-36: Basics requirements, ignition hazard assessment
- ...-37: Constructional safety “c”, control of ignition sources “b”, liquid immersion “k”
  - → *User standards on installation, inspection and maintenance, repair etc. (comparable to EN 60079-14, -17 and -19) do not yet exist*

Characteristics of Non-electrical Devices



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Risk of ignition due to devices in hazardous areas:

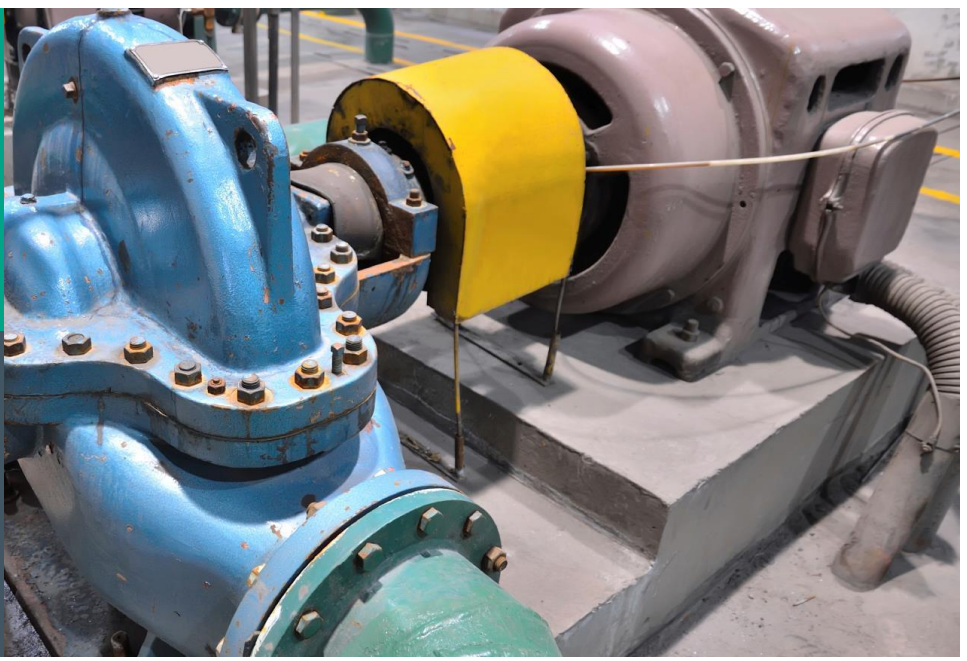
- In the case of electrical equipment, this is often already the case in normal operation
- In the case of non-electrical equipment, this is normally not the case during normal operation
  - → *Ignition hazard assessment as a basis*

Overview: Potential Ignition Sources

- Hot surfaces
- Flames and hot gases
- Mechanically generated sparks
- Electrical systems
- Electrical compensation currents and cathodic corrosion protection
- Static Electricity
- Lightning strike
- High-frequency electromagnetic waves (10^4 - 3×10^{11} Hz)
- Electromagnetic waves (3×10^{11} Hz - 3×10^{15} Hz)
- Ionizing radiation
- Ultrasound
- Adiabatic compression and shock waves
- Exothermic reactions including spontaneous combustion of dusts

2. Types of protection

c, b and k, functional principle, marking



Constructional Safety “c”



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Functional principle of constructive safety:

- No ignition sources are present in normal operation
- Risk of mechanical failures is reduced
  - → *Prevention of ignition sources becoming effective*

Protective measures "c"



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Applied protective measures:

- Ensuring an adequate level of IP protection
- Careful selection of materials, e.g. with regard to electrostatics and light metals
- Special requirements for bearings, seals, belt and chain drives, clutches, brakes, etc.

Term: Constructional Safety “c”

Ignition protection where constructional measures are applied so as to protect against the possibility of ignition from hot surfaces, sparks and adiabatic compression generated by moving parts.

EN ISO 80079-37:2016-12; 3.1

Previous Marking „c“

Previous marking acc. to EN 13463-…:

Ex II 1 G c T3

Ex II 2 G c T3

Ex II 3 G c T3



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Present Marking „c“

Present marking acc. to EN ISO 80079-…:

II 1 G Ex h IIC T3 Ga

II 2 G Ex h IIC T3 Gb

II 3 G Ex h IIC T3 Gc



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User responsibility “c”



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Reasonable care should be taken with respect to:

- Lifetimes of rolling bearings
- Required coolant or lubricant quantities
- Dry running capability
- Permissible wear limits
- Inspections

Example: Type Examination Certificate

Example: Coupling

- Category 2, non-electric: voluntary type examination

  Ex II 2 G Ex h IIC T6 … T4



IBExU Institut für Sicherheitstechnik GmbH
An-Institut der TU Bergakademie Freiberg

[1] **BAUMUSTERPRÜFBESCHEINIGUNG**

[2] für nicht-elektrische Produkte der Gerätegruppen I und II,
Gerätekategorien M2 und 2 sowie Produkte der Gerätekategorie 3

[3] Baumusterprüfbescheinigung Nummer **IBExU05ATEXB002 X** | Ausgabe 1

[4] Produkt (Gerät / Komponente):
RADEX® - NC und RADEX® - NC * HT
Servo-Lamellenkupplungen
der Bauformen DK und EK
Baugrößen jeweils bis 42 bzw. bis 61 (nur **RADEX® - NC * HT**)

a) Naben mit Passfedernut oder Spannringnaben
- 1.0 Nabe mit Passfedernut und Feststellschraube
- 2.6 Klemmnabe zweifach geschlitzt mit Passfedernut
- 3.6 Klemmnabe dreifach geschlitzt mit Passfedernut (nur **RADEX® - NC * HT**)
- 6.0 und 6.5 Nabe mit Spannring

b) Naben ohne Passfedernut
- 2.5 Klemmnabe zweifach geschlitzt ohne Passfedernut
- 3.5 Klemmnabe dreifach geschlitzt ohne Passfedernut (nur **RADEX® - NC * HT**)

An Stelle des Sterns * in RADEX® - NC * HT wird die Baugröße angegeben.

[5] Hersteller: KTR Systems GmbH

[6] Anschrift: Carl-Zeiss-Straße 25
48432 Rheine
GERMANY

[7] Dieses Produkt sowie die verschiedenen zulässigen Ausführungen sind in der Anlage zu dieser Bescheinigung sowie den darin aufgeführten Unterlagen festgelegt.

[8] IBExU Institut für Sicherheitstechnik GmbH bestätigt, dass dieses Produkt die wesentlichen Sicherheits- und Gesundheitsanforderungen für die Konzeption und den Bau von Produkten zur bestimmungsgemäßen Verwendung in explosionsgefährdeten Bereichen aus Anhang II der Richtlinie 2014/34/EU des Europäischen Parlaments und des Rates vom 26. Februar 2014 erfüllt.

Die Untersuchungs- und Prüfergebnisse werden in dem vertraulichen Prüfbericht IB-18-2-0020 festgehalten.

[9] Die Beachtung der wesentlichen Sicherheits- und Gesundheitsanforderungen wurde in Übereinstimmung mit folgenden Normen gewährleistet:
EN ISO 80079-36:2016 EN ISO 80079-37:2016 EN ISO/IEC 80079-38:2017
Hiervon ausgenommen sind jene Anforderungen, die unter Punkt [18] der Anlage aufgelistet werden.

KTR

Example: Temperature Classes

Temperature class:

Temperature class	Ambient or operating temperature T_a 1)	Max. surface temperature 2)
T4	-30 °C to +90 °C	+110 °C
T5	-30 °C to +75 °C	+95 °C
T6	-30 °C to +60 °C	+80 °C

Explanation:
The maximum surface temperatures each result from the maximum permissible ambient or operating temperature T_a plus the maximum temperature increase ΔT of 20 K to be considered. For the temperature class a safety margin subject to standard of 5 K is added.

1) The ambient or operating temperature T_a is limited to +90 °C due to the permissible permanent operating temperature of the elastomers used.
2) The maximum surface temperature of +110 °C applies for the use in locations which are potentially subject to dust explosion.

In potentially explosive atmospheres
- the ignition temperature of dusts generated must at least be 1.5 times the surface temperature to be considered
- the glow temperature must at least be the surface temperature to be considered plus a safety distance of 75 K.
- the gases and vapours generated must amount to the temperature class specified.

KTR

Example: Inspection Intervals

Equipment category	Inspection intervals
3G 3D	For couplings operated in zone 2 or zone 22 the inspection and maintenance intervals of the usual operating/assembly instructions for standard operation apply. During the standard operation which has to be taken as a basis of the ignition risk analysis the couplings are free from any ignition source. For gases, vapours and dusts generated the permissible glow and ignition temperatures specified in chapter 10.1 have to be considered and observed.
M2 2G 2D No gases and vapours of explosion group IIC	An inspection of the torsional backlash and a visual inspection of the flexible spider must be performed after 3,000 operating hours for the first time, at the latest after 6 months after start-up of the coupling. If you note insignificant or no wear on the spider upon this initial inspection, further inspections can each be performed after 6,000 operating hours or at the latest after 18 months, provided that the operating parameters remain the same. If you note significant wear during the initial inspection so that it would be recommendable to replace the spider, find out the cause according to the table „Breakdowns“, if possible. The maintenance intervals must be adjusted to the modified operating parameters without fail.
M2 2C	An inspection of the torsional backlash and a visual inspection of the flexible spider must be performed after 2,000 operating hours for the first time, at the latest after 3 months after start-up of the coupling. If you note insignificant or no wear on the spider upon this initial inspection, further

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Example: Test Criteria

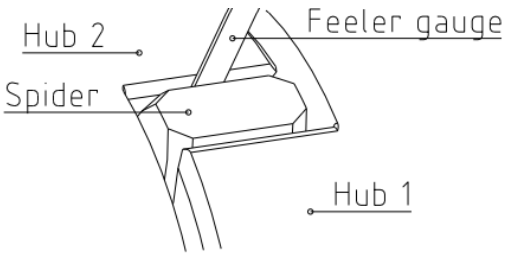


Illustration 28: Inspection of the limit of wear

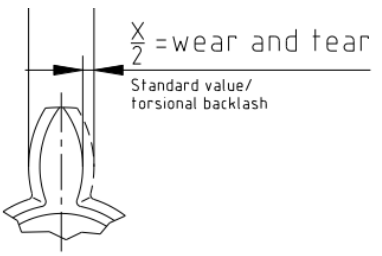


Illustration 29: Wear of spider

Table 10:

Size	Limits of wear (friction) $X_{max.}$ [mm]	Size	Limits of wear (friction) $X_{max.}$ [mm]
24	3	55	5
28	3	65	5
38	3	75	6
42	4	90	8

KTR

Example: Protective Cover

DIN EN ISO 13857.
The cover must be electrically conductive and included in the equipotential bonding. Bellhousings (magnesium share below 7.5 %) made of aluminium and damping rings (NBR) can be used as connecting element between pump and electric motor. The cover may only be taken off with standstill of the unit.

If the couplings are used in locations subject to dust explosion and in mining the user must make sure that there is no accumulation of dust in a dangerous volume between the cover and the coupling. The coupling must not operate in an accumulation of dust.

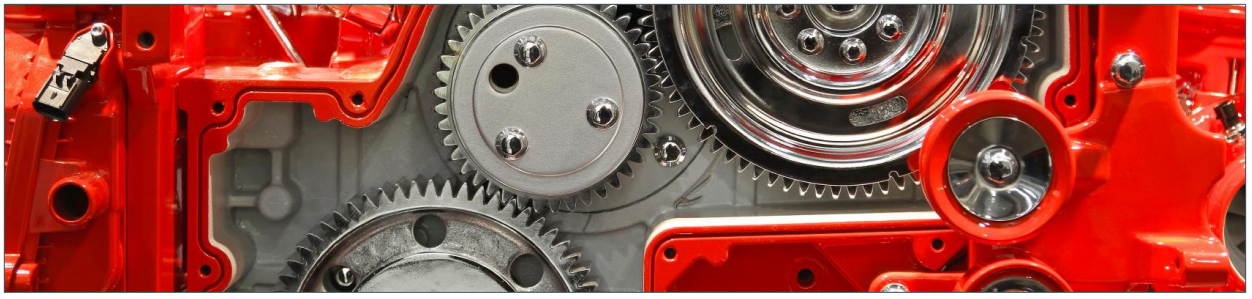
For covers with unlocked openings on the top face no light metals must be used if the couplings are used as equipment of equipment group II (*if possible, from stainless steel*). If the couplings are used in mining (equipment group I M2), the cover must not be made of light metal. In addition, it must be resistant to higher mechanical loads than with use as equipment of equipment group II.

During operation of the coupling, please pay attention to

- different operating noise
- vibrations occurring

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Liquid Immersion “k”



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Functional principle of liquid immersion:

- Cooling by protective liquid
- Spark quenching
- Separation of ignition source and explosive atmosphere

Protective Measures „k“



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Applied protective measures:

- Use of robust housing
- Use of suitable protective fluids
  - → *Protective fluid can also be used for cooling, lubricating or transferring energy.*

Term: Liquid Immersion “k”

Type of protection where potential ignition sources are made ineffective or separated from the explosive atmosphere by either totally immersing them in a protective liquid, or by partially immersing and continuously coating their active surfaces with a protective liquid in such a way that an explosive atmosphere which may be above the liquid, or outside the equipment enclosure, cannot be ignited.

EN ISO 80079-37:2016-12; 3.4

Previous Marking “k”

Previous marking acc. to EN 13463-…:

II 2 G k T3

II 3 G k T3



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Present Marking “k”

Present marking acc. to EN ISO 80079-…:

II 2 G Ex h IIC T3 Gb

II 3 G Ex h IIC T3 Gc



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User Responsibility “k”



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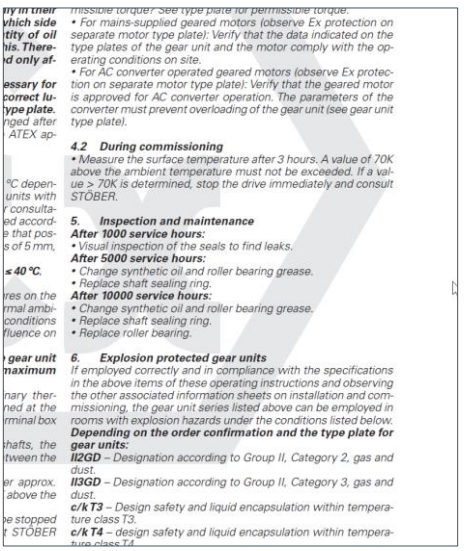
Reasonable care should be taken with respect to:

- Suitability of the protective liquid (viscosity, flammability)
- Minimum or maximum values of the protective fluid (filling level, pressure or flow rate, if applicable)
- Permissible mounting positions
- Inspections

Example: Gearbox

Operating instructions contain references to:

- Mounting position
- Surface temperatures
- Service intervals
- Etc.



missible torque? See type plate for permissible torque.
• For mains-supplied geared motors (observe Ex protection on separate motor type plate): Verify that the data indicated on the type plates of the gear unit and the motor comply with the operating conditions on site.
• For AC converter operated geared motors (observe Ex protection on separate motor type plate): Verify that the geared motor is approved for AC converter operation. The parameters of the converter must prevent overloading of the gear unit (see gear unit type plate).

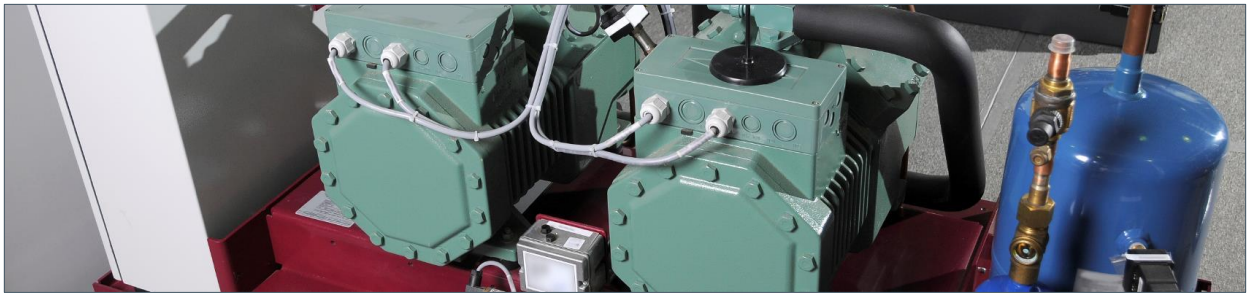
4.2 During commissioning
• Measure the surface temperature after 3 hours. A value of 70K above the ambient temperature must not be exceeded. If a value > 70K is determined, stop the drive immediately and consult STÖBER.

5. Inspection and maintenance
After 1000 service hours:
• Visual inspection of the seals to find leaks.
After 5000 service hours:
• Change synthetic oil and roller bearing grease.
• Replace shaft sealing ring.
After 10000 service hours:
• Change synthetic oil and roller bearing grease.
• Replace shaft sealing ring.
• Replace roller bearing.

6. Explosion protected gear units
If employed correctly and in compliance with the specifications in the above items of these operating instructions and observing the other associated information sheets on installation and commissioning, the gear unit series listed above can be employed in rooms with explosion hazards under the conditions listed below.
Depending on the order confirmation and the type plate for gear units:
II2GD – Designation according to Group II, Category 2, gas and dust.
II3GD – Designation according to Group II, Category 3, gas and dust.
c/k T3 – Design safety and liquid encapsulation within temperature class T3.
c/k T4 – design safety and liquid encapsulation within temperature class T4

Stöber

Control of Ignition Source “b”

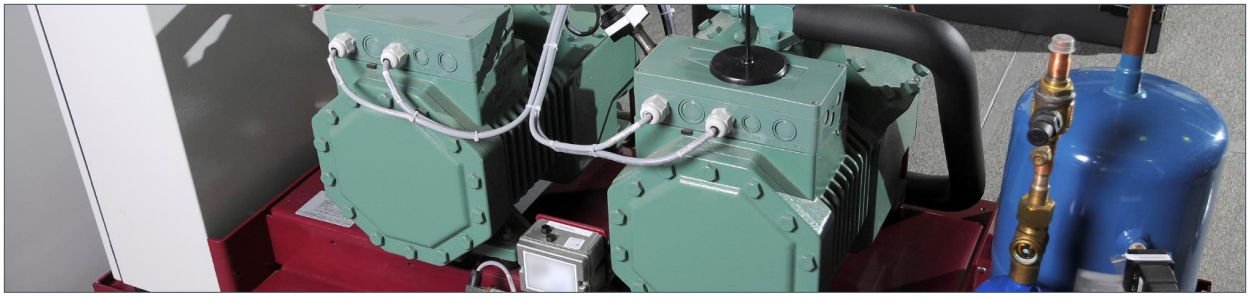


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Functional principle of the ignition source monitoring:

- Use of sensors for monitoring specified parameters
- Countermeasures for exceeding or falling short of defined target values
  - → *Example for monitoring parameters: Over temperature, over pressure or over speed, minimum quantity or flow of coolant or lubricant, etc.*

Protective Measures “b”



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Applied protective measures:

- Determination of suitable monitoring parameters (if necessary with consideration of possible response times of sensors or similar)
- Proof of the reliability of the ignition protection system

Term: Control of Ignition Source ”b”

Ignition protection where mechanical or electrical devices are used in conjunction with non-electrical equipment to manually or automatically reduce the likelihood of a potential ignition source from becoming an effective ignition source.

EN ISO 80079-37:2016-12; 3.3

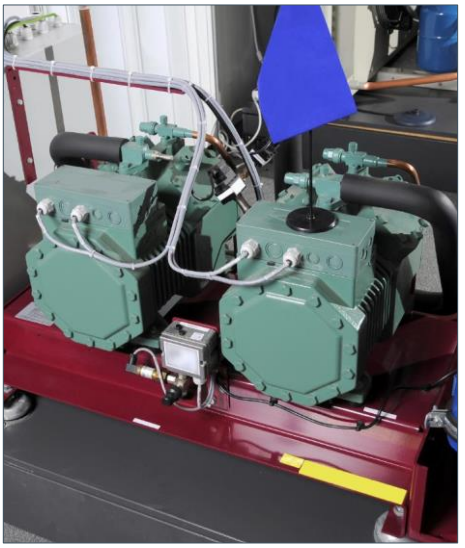
Previous Marking "b”

Previous marking acc. to EN 13463-...:

II 1 G b T3

II 2 G b T3

II 3 G b T3



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Present Marking "b”

Present marking acc. to EN ISO 80079-...:

II 1 G Ex h IIC T3 Ga

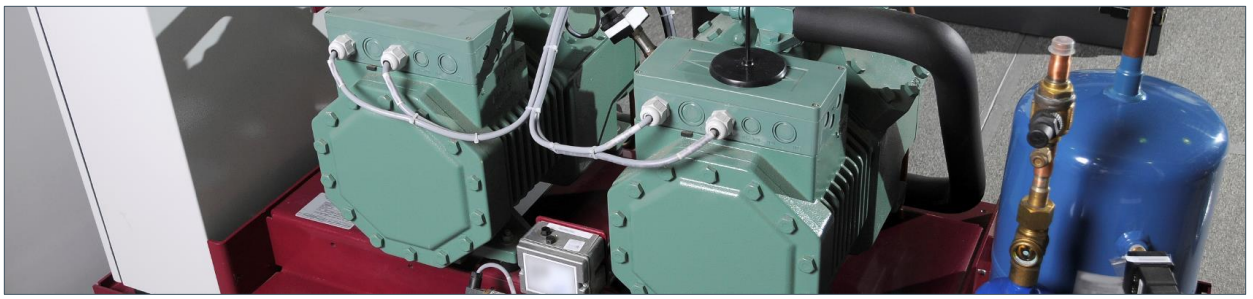
II 2 G Ex h IIC T3 Gb

II 3 G Ex h IIC T3 Gc



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User Responsibility "b"



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Reasonable care should be taken with respect to:

- Differentiation between ignition protection systems type b1 and b2
- Inspections

Ignition Prevention System Type b1

Requirements:

- Based on components with an appropriate level of reliability
- Conforms to relevant standards
- Application of proven security principles
- Suitable for expected operating and environmental influences

Characteristics:

- If set target values are exceeded or fallen short of: automatic action or warning signal
- Inspections of the ignition protection system in an appropriate manner

Ignition Prevention System Type b2

Requirements:

- Based on components with an appropriate level of reliability
- Conforms to relevant standards
- Application of proven security principles
- Suitable for expected operating and environmental influences

Characteristics:

- If set target values are exceeded or fallen short of: automatic action or ~~warning signal~~
- Inspections of the ignition protection system in an appropriate manner

Other Types of Protection

Other types of protection types of the earlier series EN 13463-...:

- Flameproof enclosure “d”
- Restricted breathing enclosure “fr”



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